ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** outside the circle	Compare 9 cm with the radius 5 cm.
2	**(c)** 4	Midpoint of AB.
3	**(b)** 5	Diagonals of a rectangle are equal and bisect each other.
4	**(d)** 60	Triangle OAB is isosceles with OA = OB.
5	**(a)** 4	Subtract the radii.
6	**(d)** equal and perpendicular bisectors of each other	Property
7	**(a)** 2	Two pairs of equal opposite sides.
8	**(d)** 30	$2(l + b)$.
9	**(b)** outside the circle	Compare 8 cm with the radius 5 cm.
10	**(c)** 7	Midpoint of AB.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Compare 4 cm with the radius 6 cm (1); P lies inside the circle (1).	1+1
12	Diagonals are equal and bisect each other (1); OA = OB = 6 cm; BD = 12 cm (1).	1+1
13	Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 3 cm from A (1).	1+1
14	Compare 7 cm with the radius 7 cm (1); P lies on the circle (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Draw AB = 6 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 3 cm from A (1).	1+1+1
16	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
17	Diameter $= 8$ cm (1); Longest chord $= 8$ cm (1); No: 9 cm exceeds the diameter (1).	1+1+1
	OR: Draw AB = 7 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 3.5 cm from A (1).	1+1+1
18	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
	OR: Diameter $= 12$ cm (1); Longest chord $= 12$ cm (1); No: 13 cm exceeds the diameter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 8 cm (1); Draw perpendiculars at A and B and mark 6 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: P: inside (1); Q: on (1); R: outside (1); Inside or on the circle: 2 (2).	1+1+1+2
20	Draw AB = 12 cm (1); Draw perpendiculars at A and B and mark 5 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: P: on (1); Q: outside (1); R: inside (1); Inside or on the circle: 2 (2).	1+1+1+2

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.

- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 8	Diagonals of a rectangle are equal and bisect each other.
2	**(b)** 50	Triangle OAB is isosceles with OA = OB.
3	**(d)** 4	Subtract the radii.
4	**(b)** are equal but not perpendicular	Property
5	**(d)** 6	It equals the side.
6	**(c)** 44	$2(l + b)$.
7	**(c)** inside the circle	Compare 4 cm with the radius 6 cm.
8	**(b)** 9	Midpoint of AB.
9	**(c)** 10	Diagonals of a rectangle are equal and bisect each other.
10	**(b)** 40	Triangle OAB is isosceles with OA = OB.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Diagonals are equal and bisect each other (1); OA = OB = 7 cm; BD = 14 cm (1).	1+1
12	Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 5 cm from A (1).	1+1
13	Compare 9 cm with the radius 5 cm (1); P lies outside the circle (1).	1+1
14	Diagonals are equal and bisect each other (1); OA = OB = 9 cm; BD = 18 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Draw AB = 8 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 4 cm from A (1).	1+1+1
16	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
17	Diameter $= 16$ cm (1); Longest chord $= 16$ cm (1); No: 17 cm exceeds the diameter (1).	1+1+1
	OR: Draw AB = 9 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 4.5 cm from A (1).	1+1+1
18	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
	OR: Diameter $= 20$ cm (1); Longest chord $= 20$ cm (1); No: 21 cm exceeds the diameter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 6 cm (1); Draw perpendiculars at A and B and mark 4 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: P: on (1); Q: inside (1); R: outside (1); Inside or on the circle: 2 (2).	1+1+1+2
20	Draw AB = 15 cm (1); Draw perpendiculars at A and B and mark 8 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: P: outside (1); Q: on (1); R: inside (1); Inside or on the circle: 2 (2).	1+1+1+2

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.

- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 7	Subtract the radii.
2	**(a)** perpendicular bisector of AB	Property
3	**(d)** 8	It equals the side.
4	**(a)** outside the circle	Compare 5 cm with the radius 4 cm.
5	**(c)** 12	Midpoint of AB.
6	**(b)** 13	Diagonals of a rectangle are equal and bisect each other.
7	**(b)** 30	Triangle OAB is isosceles with OA = OB.
8	**(d)** 4	Subtract the radii.
9	**(b)** 10	It equals the side.
10	**(b)** inside the circle	Compare 2 cm with the radius 7 cm.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 8 cm from A (1).	1+1
12	Compare 3 cm with the radius 8 cm (1); P lies inside the circle (1).	1+1
13	Diagonals are equal and bisect each other (1); OA = OB = 11 cm; BD = 22 cm (1).	1+1
14	Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 10 cm from A (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Draw AB = 10 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 5 cm from A (1).	1+1+1
16	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
17	Diameter $= 24$ cm (1); Longest chord $= 24$ cm (1); No: 25 cm exceeds the diameter (1).	1+1+1
	OR: Draw AB = 12 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 6 cm from A (1).	1+1+1
18	Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1).	1+1+1
	OR: Diameter $= 30$ cm (1); Longest chord $= 30$ cm (1); No: 31 cm exceeds the diameter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 9 cm (1); Draw perpendiculars at A and B and mark 4 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: P: outside (1); Q: on (1); R: inside (1); Inside or on the circle: 2 (2).	1+1+1+2
20	Draw AB = 7 cm (1); Draw perpendiculars at A and B and mark 5 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1).	1+1+1+1+1
	OR: P: inside (1); Q: on (1); R: outside (1); Inside or on the circle: 2 (2).	1+1+1+2

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.

- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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